

2303A52416

batch=35

Task Description #1 – Stack Implementation

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample Input Code:

class Stack:

pass

Expected Output:

- A functional stack implementation with all required methods and docstrings.

Code

class Stack:

"""

Stack Data Structure (LIFO - Last In First Out)

Methods:

push(item) -> Add item to top of stack

pop() -> Remove and return top item

peek() -> Return top item without removing

is_empty() -> Check if stack is empty

"""

def __init__(self):

"""Initialize an empty stack."""

self.items = []

def push(self, item):

"""Add an item to the top of the stack."""

self.items.append(item)

def pop(self):

"""Remove and return the top item from the stack."""

if not self.is_empty():

return self.items.pop()

return "Stack is empty"

def peek(self):

"""Return the top item without removing it."""

if not self.is_empty():

return self.items[-1]

return "Stack is empty"

def is_empty(self):

"""Return True if stack is empty, else False."""

return len(self.items) == 0

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```
main.py
1 class Stack:
2     """
3     Stack Data Structure (LIFO - Last In First Out)
4
5     Methods:
6     push(item) -> Add item to top of stack
7     pop()      -> Remove and return top item
8     peek()     -> Return top item without removing
9     is_empty() -> Check if stack is empty
10    """
11
12    def __init__(self):
13        """Initialize an empty stack."""
14        self.items = []
15
16    def push(self, item):
17        """Add an item to the top of the stack."""
18        self.items.append(item)
19
20    def pop(self):
21        """Remove and return the top item from the stack."""
22        if not self.is_empty():
23            return self.items.pop()
24        return "Stack is empty"
25
26    def peek(self):
27        """Return the top item without removing it."""
28        if not self.is_empty():
29            return self.items[-1]
30        return "Stack is empty"
31
32    def is_empty(self):
33        """Return True if stack is empty, else False."""
34        return len(self.items) == 0
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```

```
class Queue:
```

```
pass
```

Expected Output:

- FIFO-based queue class with enqueue, dequeue, peek, and size methods.

Code

```
class Queue:
```

```
    """
```

```
    Queue Data Structure (FIFO - First In First Out)
```

Methods:

```
    enqueue(item) -> Add item to the rear of the queue
```

```
    dequeue()     -> Remove and return the front item
```

```
    peek()        -> Return the front item without removing it
```

```
    size()        -> Return number of elements in the queue
```

```
    """
```

```
def __init__(self):
```

```
    """Initialize an empty queue."""
```

```
    self.items = []
```

```
def enqueue(self, item):
```

```
    """
```

```
    Add an item to the rear of the queue.
```

```
    """
```

```
    self.items.append(item)
```

```
def dequeue(self):
```

```
    """
```

```
    Remove and return the front item of the queue.
```

```
    Raises IndexError if queue is empty.
```

```
    """
```

```
    if self.size() == 0:
```

```
        raise IndexError("Dequeue from empty queue")
```

```
    return self.items.pop(0)
```

```
def peek(self):
```

```
    """
```

```
    Return the front item without removing it.
```

```
    Raises IndexError if queue is empty.
```

```
    """
```

```
    if self.size() == 0:
```

```
        raise IndexError("Peek from empty queue")
```

```
    return self.items[0]
```

```
def size(self):
```

```
    """
```

```
    Return the number of elements in the queue.
```

```
"""
```

```
return len(self.items)
```

```
def is_empty(self):
```

```
    """
```

```
    Return True if queue is empty, otherwise False.
```

```
    """
```

```
    return self.size() == 0
```

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Run

Debug

Stop

Share

Save

Beautify

Language Python 3

```
1- class Queue:
2-     """
3-     Queue Data Structure (FIFO - First In First Out)
4-
5-     Methods:
6-         enqueue(item) -> Add item to the rear of the queue
7-         dequeue()     -> Remove and return the front item
8-         peek()        -> Return the front item without removing it
9-         size()        -> Return number of elements in the queue
10-    """
11-
12-    def __init__(self):
13-        """Initialize an empty queue."""
14-        self.items = []
15-
16-    def enqueue(self, item):
17-        """
18-        Add an item to the rear of the queue.
19-        """
20-        self.items.append(item)
21-
22-    def dequeue(self):
23-        """
24-        Remove and return the front item of the queue.
```

input

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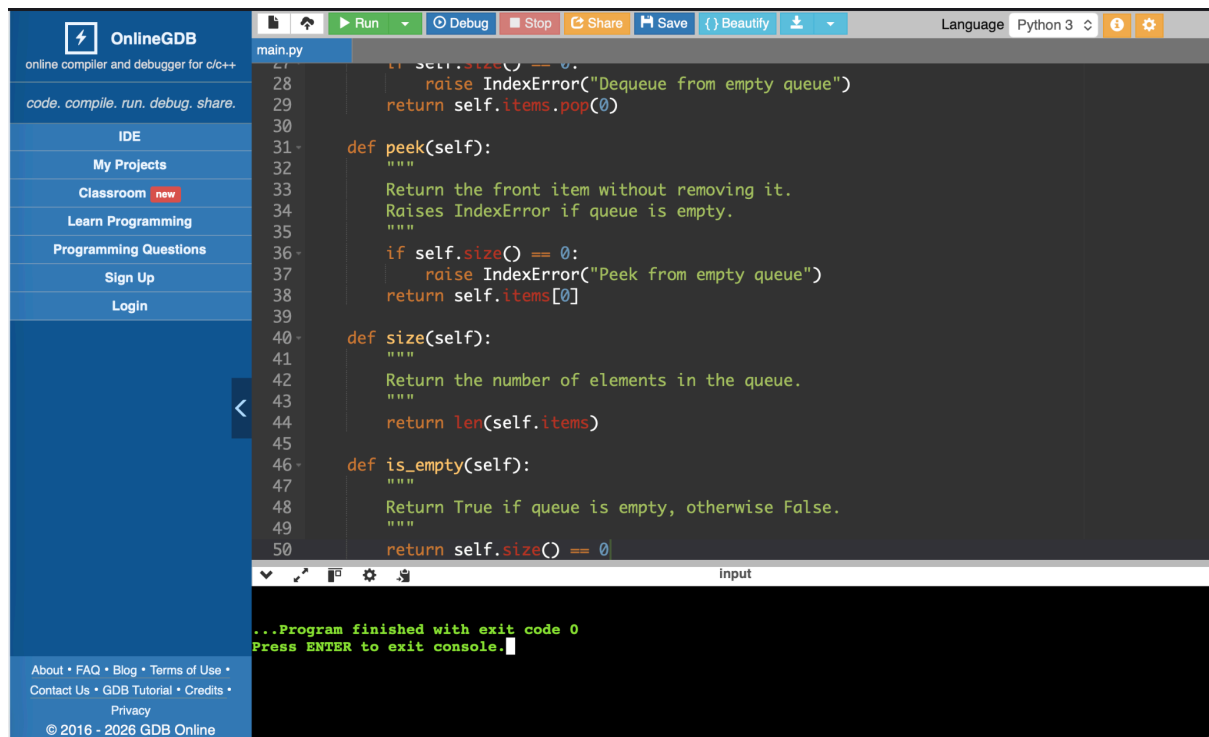
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Language Python 3

```
23-
24-     Remove and return the front item of the queue.
25-     Raises IndexError if queue is empty.
26-     """
27-
28-     if self.size() == 0:
29-         raise IndexError("Dequeue from empty queue")
30-     return self.items.pop(0)
31-
32-    def peek(self):
33-        """
34-        Return the front item without removing it.
35-        Raises IndexError if queue is empty.
36-        """
37-
38-        if self.size() == 0:
39-            raise IndexError("Peek from empty queue")
40-        return self.items[0]
41-
42-    def size(self):
43-        """
44-        Return the number of elements in the queue.
45-        """
46-        return len(self.items)
47-
48-    def is_empty(self):
```

input

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Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and display methods.

Sample Input Code:

class Node:

pass

class LinkedList:

pass

Expected Output:

- A working linked list implementation with clear method documentation.

Code

class Node:

"""Node class for Linked List."""

def __init__(self, data):

self.data = data

self.next = None

class LinkedList:

"""

Singly Linked List

Methods:

insert(data) -> Insert at end

display() -> Display list elements

```
"""
```

```
def __init__(self):
    self.head = None

def insert(self, data):
    """Insert new node at end of list."""
    new_node = Node(data)

    if self.head is None:
        self.head = new_node
        return

    temp = self.head
    while temp.next:
        temp = temp.next

    temp.next = new_node

def display(self):
    """Print all elements in the list."""
    temp = self.head
    while temp:
        print(temp.data, end=" -> ")
        temp = temp.next
    print("None")
```

Explanation

Follows FIFO principle.

append() → add at rear

pop(0) → remove from front

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```

1 class Node:
2     """Node class for Linked List."""
3
4     def __init__(self, data):
5         self.data = data
6         self.next = None
7
8
9 class LinkedList:
10     """
11     Singly Linked List
12
13     Methods:
14     insert(data) -> Insert at end
15     display() -> Display list elements
16     """
17
18     def __init__(self):
19         self.head = None
20
21     def insert(self, data):
22         """Insert new node at end of list."""
23         new_node = Node(data)

```

input

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```

18 def __init__(self):
19     self.head = None
20
21 def insert(self, data):
22     """Insert new node at end of list."""
23     new_node = Node(data)
24
25     if self.head is None:
26         self.head = new_node
27         return
28
29     temp = self.head
30     while temp.next:
31         temp = temp.next
32
33     temp.next = new_node
34
35 def display(self):
36     """Print all elements in the list."""
37     temp = self.head
38     while temp:
39         print(temp.data, end=" -> ")
40         temp = temp.next
41     print("None")

```

input

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Press ENTER to exit console.

Task Description #4 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Sample Input Code:

```
class HashTable:
```

```
pass
```

Expected Output:

- Collision handling using chaining, with well-commented methods.

Code


```

class HashTable:
    """
    Hash Table with Separate Chaining

    Methods:
    insert(key, value)
    search(key)
    delete(key)
    """

    def __init__(self, size=10):
        self.size = size
        self.table = [[] for _ in range(size)]

    def hash_function(self, key):
        """Generate index using built-in hash."""
        return hash(key) % self.size

    def insert(self, key, value):
        """Insert key-value pair into hash table."""
        index = self.hash_function(key)
        self.table[index].append((key, value))

    def search(self, key):
        """Search value by key."""
        index = self.hash_function(key)
        for k, v in self.table[index]:
            if k == key:
                return v
        return "Key not found"

    def delete(self, key):
        """Delete key-value pair."""
        index = self.hash_function(key)
        for i, (k, v) in enumerate(self.table[index]):
            if k == key:
                del self.table[index][i]
                return "Deleted"
        return "Key not found"

```

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```

1 class HashTable:
2     """
3     Hash Table with Separate Chaining
4
5     Methods:
6     insert(key, value)
7     search(key)
8     delete(key)
9     """
10
11 def __init__(self, size=10):
12     self.size = size
13     self.table = [[] for _ in range(size)]
14
15 def hash_function(self, key):
16     """Generate index using built-in hash."""
17     return hash(key) % self.size
18
19 def insert(self, key, value):
20     """Insert key-value pair into hash table."""
21     index = self.hash_function(key)
22     self.table[index].append((key, value))
23
24 def search(self, key):

```

input

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```

16     """Generate index using built-in hash.
17     return hash(key) % self.size
18
19 def insert(self, key, value):
20     """Insert key-value pair into hash table."""
21     index = self.hash_function(key)
22     self.table[index].append((key, value))
23
24 def search(self, key):
25     """Search value by key."""
26     index = self.hash_function(key)
27     for k, v in self.table[index]:
28         if k == key:
29             return v
30     return "Key not found"
31
32 def delete(self, key):
33     """Delete key-value pair."""
34     index = self.hash_function(key)
35     for i, (k, v) in enumerate(self.table[index]):
36         if k == key:
37             del self.table[index][i]
38             return "Deleted"
39     return "Key not found"

```

input

...Program finished with exit code 0
Press ENTER to exit console.

Explanation

Uses separate chaining for collision handling.

Hash function maps key to bucket index.

Each bucket is a list of key-value pairs.

Task Description #5 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

```
class Graph:
```

pass

Expected Output:

- Graph with methods to add vertices, add edges, and display connections.

Code

class Graph:

"""

Graph Representation using Adjacency List

Methods:

add_vertex(vertex)

add_edge(v1, v2)

display()

"""

def __init__(self):

self.adj_list = {}

def add_vertex(self, vertex):

"""Add new vertex to graph."""

if vertex not in self.adj_list:

self.adj_list[vertex] = []

def add_edge(self, v1, v2):

"""Add edge between two vertices."""

if v1 in self.adj_list and v2 in self.adj_list:

self.adj_list[v1].append(v2)

self.adj_list[v2].append(v1)

def display(self):

"""Display adjacency list."""

for vertex in self.adj_list:

print(vertex, "->", self.adj_list[vertex])

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main.py

```
1 class Graph:
2     """
3     Graph Representation using Adjacency List
4
5     Methods:
6     add_vertex(vertex)
7     add_edge(v1, v2)
8     display()
9     """
10
11 def __init__(self):
12     self.adj_list = {}
13
14 def add_vertex(self, vertex):
15     """Add new vertex to graph."""
16     if vertex not in self.adj_list:
17         self.adj_list[vertex] = []
18
19 def add_edge(self, v1, v2):
20     """Add edge between two vertices."""
21     if v1 in self.adj_list and v2 in self.adj_list:
22         self.adj_list[v1].append(v2)
23         self.adj_list[v2].append(v1)
24
25
26 ...Program finished with exit code 0
27 Press ENTER to exit console.
```

input

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main.py

```
5 Methods:
6 add_vertex(vertex)
7 add_edge(v1, v2)
8 display()
9 """
10
11 def __init__(self):
12     self.adj_list = {}
13
14 def add_vertex(self, vertex):
15     """Add new vertex to graph."""
16     if vertex not in self.adj_list:
17         self.adj_list[vertex] = []
18
19 def add_edge(self, v1, v2):
20     """Add edge between two vertices."""
21     if v1 in self.adj_list and v2 in self.adj_list:
22         self.adj_list[v1].append(v2)
23         self.adj_list[v2].append(v1)
24
25 def display(self):
26     """Display adjacency list."""
27     for vertex in self.adj_list:
28         print(vertex, "->", self.adj_list[vertex])
29
30
31 ...Program finished with exit code 0
32 Press ENTER to exit console.
```

input

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Explanation

Dictionary stores adjacency list.

Efficient for sparse graphs.

Suitable for networks.

Structure Selection

A hospital wants to develop a Smart Hospital Management System that handles:

1. Patient Check-In System – Patients are registered and treated in order of arrival.
2. Emergency Case Handling – Critical patients must be treated first.
3. Medical Records Storage – Fast retrieval of patient details using ID.
4. Doctor Appointment Scheduling – Appointments sorted by time.
5. Hospital Room Navigation – Represent connections between wards and rooms.

Student Task

- For each feature, select the most appropriate data structure from the list below:
 - o Stack
 - o Queue
 - o Priority Queue
 - o Linked List
 - o Binary Search Tree (BST)
 - o Graph
 - o Hash Table
 - o Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

Code

```
import heapq
```

```
class EmergencySystem:
```

```
    """
```

```
    Emergency Patient Management System using Priority Queue.
```

```
    Lower priority number = Higher urgency.
```

```
    Example:
```

```
        Priority 1 -> Critical
```

```
        Priority 2 -> Serious
```

```
        Priority 3 -> Normal
```

```
    """
```

```
    def __init__(self):
```

```
        """Initialize an empty emergency queue."""
```

```
        self.patients = []
```

```
    def add_patient(self, name, priority):
```

```
        """
```

```
        Add a patient to the emergency queue.
```

```

:param name: Name of the patient
:param priority: Urgency level (lower number = higher priority)
"""

heapq.heappush(self.patients, (priority, name))
print(f"Patient {name} added with priority {priority}")

def treat_patient(self):
    """
    Treat the highest priority patient.
    :return: Patient details
    """
    if self.patients:
        priority, name = heapq.heappop(self.patients)
        return f"Treating {name} (Priority {priority})"
    return "No patients in emergency queue."

def display_patients(self):
    """
    Display all waiting patients.
    """
    if not self.patients:
        print("No patients waiting.")
    else:
        print("Current Emergency Queue:")
        for patient in self.patients:
            print(f"Priority {patient[0]} - {patient[1]}")

```

```
1 import heapq
2
3 class EmergencySystem:
4     """
5     Emergency Patient Management System using Priority Queue.
6
7     Lower priority number = Higher urgency.
8     Example:
9         Priority 1 -> Critical
10        Priority 2 -> Serious
11        Priority 3 -> Normal
12    """
13
14    def __init__(self):
15        """Initialize an empty emergency queue."""
16        self.patients = []
17
18    def add_patient(self, name, priority):
19        """
20        Add a patient to the emergency queue.
21
22        :param name: Name of the patient
23        :param priority: Urgency level (Lower number = higher priority)
24    """
```

...Program finished with exit code 0
Press ENTER to exit console.

```
24
25     heapq.heappush(self.patients, (priority, name))
26     print(f"Patient {name} added with priority {priority}")
27
28     def treat_patient(self):
29         """
30         Treat the highest priority patient.
31         :return: Patient details
32         """
33         if self.patients:
34             priority, name = heapq.heappop(self.patients)
35             return f"Treating {name} (Priority {priority})"
36             return "No patients in emergency queue."
37
38     def display_patients(self):
39         """
40         Display all waiting patients.
41         """
42         if not self.patients:
43             print("No patients waiting.")
44         else:
45             print("Current Emergency Queue:")
46             for patient in self.patients:
47                 print(f"Priority {patient[0]} - {patient[1]}")
```

...Program finished with exit code 0
Press ENTER to exit console.

ask Description #7: Smart City Traffic Control System

A city plans a Smart Traffic Management System that includes:

1. Traffic Signal Queue – Vehicles waiting at signals.
2. Emergency Vehicle Priority Handling – Ambulances and fire trucks prioritized.
3. Vehicle Registration Lookup – Instant access to vehicle details.
4. Road Network Mapping – Roads and intersections connected logically.
5. Parking Slot Availability – Track available and occupied slots.

Student Task

- For each feature, select the most appropriate data structure from the list below:
 - o Stack
 - o Queue
 - o Priority Queue
 - o Linked List
 - o Binary Search Tree (BST)
 - o Graph
 - o Hash Table
 - o Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

Code

```
class VehicleRegistry:
    """
    Vehicle Registration using Hash Table
    """

    def __init__(self):
        self.registry = {}

    def register(self, vehicle_no, owner):
        """Register vehicle."""
        self.registry[vehicle_no] = owner

    def lookup(self, vehicle_no):
        """Lookup vehicle details."""
        return self.registry.get(vehicle_no, "Not found")
```



```
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15        return self.registry.get(vehicle_no, "Not found")
```

...Program finished with exit code 0
Press ENTER to exit console.

```
class OrderSystem:
    """
    Order Processing using Queue
    """
```

```
def __init__(self):
    self.orders = []

def place_order(self, order_id):
    """Add new order."""
    self.orders.append(order_id)

def process_order(self):
    """Process first order."""
    if self.orders:
        return self.orders.pop(0)
    return "No orders"
```

Explanation

The queue follows the FIFO (First In First Out) principle.

- enqueue() → adds elements at the end using append().
- dequeue() → removes elements from the front using pop(0).
- peek() → checks the first element without removing it.
- size() → returns the total number of elements.

⌚ Time Complexity

- enqueue() → $O(1)$
- dequeue() → $O(n)$ (because pop(0) shifts elements)

peek() $\rightarrow O(1)$
size() $\rightarrow O(1)$